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SUBJECT: CLOUD COMPUTING LAB

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Section: 5-B

LAB NO 10:

Task 1 — GitHub CLI Codespace Setup & Authentication

1. Install GitHub CLI:

```
PS C:\Users\saira> winget install --id GitHub.cli
Found an existing package already installed. Trying to upgrade the installed package...
Found GitHub CLI [GitHub.cli] Version 2.83.2
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/cli/cli/releases/download/v2.83.2/gh_2.83.2_windows_amd64.msi
17.7 MB / 17.7 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\saira>
```

AKIAVHLXGUTLFHO3WLDL

lyrOoL8RGFoke93ualPPII3iV5Bxy0nmDOHFT4FB

2. Authenticate GH CLI for Codespaces:

```
PS C:\Users\saira> gh auth login -s codespace
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Paste an authentication token
Tip: you can generate a Personal Access Token here https://github.com/settings/tokens
The minimum required scopes are 'repo', 'read:org', 'workflow'.
? Paste your authentication token: *****
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as 23-22411-057-sudo
! You were already logged in to this account
PS C:\Users\saira>
```

3. List available codespaces:

```
PS C:\Users\saira> gh codespace list
NAME              DISPLAY_NAME  REPOSITORY      BRANCH  STATE    CREATED_AT
studious-fishstick-r454rprpjx94f6p7  studious fishstick  23-22411-057-sudo/exam_...  main*   Shutdown  about 21 days ago
PS C:\Users\saira> |
```

4. Connect to a codespace via SSH:

```
View
PS C:\Users\saira> gh codespace view studious-fishstick-r454rprpjx94f6p7
the command accepts no arguments
PS C:\Users\saira> gh codespace ssh -c studious-fishstick-r454rprpjx94f6p7
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
Last login: Sat Dec  6 06:22:34 2025 from ::1
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

Task 2 — Install AWS CLI, Terraform CLI, Provider Setup

1. In Codespace shell, install AWS CLI:

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
```

```
unzip awscliv2.zip
```

```
sudo ./aws/install
```

```
aws --version
```

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
unzip awscliv2.zip
sudo ./aws/install
aws --version
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 60.3M  100 60.3M    0     0  194M      0  --:--:-- --:--:-- --:--:-- 195M
Archive:  awscliv2.zip
replace aws/install? [y]es, [n]o, [A]ll, [N]one, [r]ename: yes
inflating: aws/install
replace aws/THIRD_PARTY_LICENSES? [y]es, [n]o, [A]ll, [N]one, [r]ename: yes
inflating: aws/THIRD_PARTY_LICENSES
replace aws/README.md? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
inflating: aws/README.md
inflating: aws/dist/aws
inflating: aws/dist/aws_completer
inflating: aws/dist/libpython3.13.so.1.0
inflating: aws/dist/_awscli.abi3.so
inflating: aws/dist/_ruamel.yaml.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/libz.so.1
inflating: aws/dist/liblzma.so.5
inflating: aws/dist/libbz2.so.1
inflating: aws/dist/libbrotli.so.1
inflating: aws/dist/libuuid.so.1
inflating: aws/dist/libreadline.so.8
inflating: aws/dist/libtinfo.so.6
inflating: aws/dist/libsqlite3.so.0
inflating: aws/dist/base_library.zip
inflating: aws/dist/lib-dynload/_datetime.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/unicodedata.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_csv.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_statistics.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_contextvars.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_decimal.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_pickle.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_hashlib.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_sha3.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_blake2.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_md5.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_sha1.cpython-313-x86_64-linux-gnu.so
inflating: aws/dist/lib-dynload/_sha2.cpython-313-x86_64-linux-gnu.so
```

```

inflating: aws/dist/awscli/examples/health/describe-event-details.rst
inflating: aws/dist/awscli/examples/health/describe-affected-entities.rst
inflating: aws/dist/awscli/examples/serverlessrepo/put-application-policy.rst
inflating: aws/dist/awscli/examples/snowball/get-snowball-usage.rst
inflating: aws/dist/awscli/examples/snowball/list-jobs.rst
inflating: aws/dist/awscli/examples/maciek2/describe-buckets.rst
inflating: aws/dist/awscli/examples/dsql/generate-db-connect-auth-token.rst
inflating: aws/dist/awscli/topics/s3-faq.rst
inflating: aws/dist/awscli/topics/s3-case-insensitivity.rst
inflating: aws/dist/awscli/topics/config-vars.rst
inflating: aws/dist/awscli/topics/return-codes.rst
inflating: aws/dist/awscli/topics/topic-tags.json
inflating: aws/dist/awscli/topics/s3-config.rst
inflating: aws/dist/awscli/topics/ddb-expressions.rst
inflating: aws/dist/awscli/customizations/sso/index.html
inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
inflating: aws/dist/awscli/data/cli.json
inflating: aws/dist/awscli/data/ac.index
inflating: aws/dist/awscli/data/metadata.json
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflating: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflating: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflating: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflating: aws/dist/wheel-0.45.1.dist-info/METADATA
inflating: aws/dist/wheel-0.45.1.dist-info/RECORD
inflating: aws/dist/wheel-0.45.1.dist-info/WHEEL
inflating: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
ound preexisting AWS CLI installation: /usr/local/aws-cli/v2/current. Please rerun install script with --update flag.
aws-cli/2.32.10 Python/3.13.9 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
23-22411-057-sudo → /workspaces/exam_prep (main) $

```

2. Configure AWS CLI:

aws configure

```

AWS Secret Access Key [AKIAVHXLXGUTLFHO3WLDL]:
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws configure
AWS Access Key ID [AKIAVHXLXGUTLFHO3WLDL]: AKIAVHXLXGUTLFHO3WLDL
AWS Secret Access Key [AKIAVHXLXGUTLFHO3WLDL]: lyr0oL8RGFoke93ualPPII3iV5Bxy0nmDOHFT4FB
Default region name [me-central-1]: me-central-1
Default output format [json]: json
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```

3. Then check config files:

```
cat ~/.aws/credentials
```

```
cat ~/.aws/config
```

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat ~/.aws/credentials
[default]
aws_access_key_id = AKIAVHXLXGUTLFHO3WLDL
aws_secret_access_key = lyr0oL8RGFoke93ualPPII3iV5Bxy0nmDOHFT4FB
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat ~/.aws/config
[default]
region = me-central-1
output = json
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```

4. Verify AWS CLI connectivity:

aws sts get-caller-identity

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws sts get-caller-identity
{
  "UserId": "AIDAVHXLXGUTLK25C57B4F",
  "Account": "359416636630",
  "Arn": "arn:aws:iam::359416636630:user/Admin"
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```

B. Install Terraform CLI

1. Install Terraform CLI:

```
wget -O - https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/hashicorp-archive-keyring.gpg echo "deb [arch=$(dpkg --print-architecture)
signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com
$(grep -oP '(?<=UBUNTU_CODENAME=).*' /etc/os-release || lsb_release -cs) main" | sudo tee
/etc/apt/sources.list.d/hashicorp.list
```

```
sudo apt update
```

```
sudo apt install terraform
```

```
which terraform
```

```
terraform --version
```

```
823-22411-057-sudo → /workspaces/exam_prep (main) $ wget -O - https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-a
rchive-keyring.gpg
--2025-12-26 11:30:30-- https://apt.releases.hashicorp.com/gpg
Resolving apt.releases.hashicorp.com (apt.releases.hashicorp.com)... 18.172.78.129, 18.172.78.30, 18.172.78.65, ...
Connecting to apt.releases.hashicorp.com (apt.releases.hashicorp.com)|18.172.78.129|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3980 (3.9K) [binary/octet-stream]
Saving to: 'STDOUT'

-
100%[=====] 3.89K --.-KB/s in 0s

2025-12-26 11:30:30 (713 MB/s) - written to stdout [3980/3980]

823-22411-057-sudo → /workspaces/exam_prep (main) $ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]
https://apt.releases.hashicorp.com $(grep -oP '(?<=UBUNTU_CODENAME=).*' /etc/os-release || lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/has
hi
corp.list
deb [arch=amd64 signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com noble main
823-22411-057-sudo → /workspaces/exam_prep (main) $ sudo apt update
Get:1 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Get:2 https://dl.yarnpkg.com/debian stable InRelease
Get:3 https://repo.anaconda.com/pkgs/misc/debrepo/conda stable InRelease [3961 B]
Get:4 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble InRelease [3600 B]
Get:5 https://repo.anaconda.com/pkgs/misc/debrepo/conda stable/main amd64 Packages [4557 B]
Get:6 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main amd64 Packages [77.1 kB]
Get:7 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main all Packages [643 B]
Get:8 https://dl.yarnpkg.com/debian stable/main amd64 Packages [11.8 kB]
Get:9 https://dl.yarnpkg.com/debian stable/main all Packages [11.8 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:12 https://apt.releases.hashicorp.com noble/main amd64 Packages [266 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1752 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1183 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2898 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3059 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1183 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2898 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3059 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2130 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1950 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [35.9 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.5 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [34.6 kB]
Fetched 35.7 MB in 4s (8060 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
68 packages can be upgraded. Run 'apt list --upgradable' to see them.
823-22411-057-sudo → /workspaces/exam_prep (main) $ sudo apt install terraform
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  terraform
0 upgraded, 1 newly installed, 0 to remove and 68 not upgraded.
Need to get 30.6 MB of archives.
After this operation, 101 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com noble/main amd64 terraform amd64 1.14.3-1 [30.6 MB]
Fetched 30.6 MB in 0s (163 MB/s)
Selecting previously unselected package terraform.
(Reading database ... 58631 files and directories currently installed.)
Preparing to unpack .../terraform_1.14.3-1_amd64.deb ...
Unpacking terraform (1.14.3-1) ...
Setting up terraform (1.14.3-1) ...
823-22411-057-sudo → /workspaces/exam_prep (main) $ which terraform
/usr/bin/terraform
823-22411-057-sudo → /workspaces/exam_prep (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
```

C. Provider Configuration (main.tf)

1. Create main.tf using Vim:

```
vim main.tf
```



```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ vim main.tf
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

2. Inside main.tf, write:

```
provider "aws" {

  shared_config_files    = ["~/.aws/config"]

  shared_credentials_files = ["~/.aws/credentials"]

}
```

```
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
~
~
~
~
~
~
~
~
```

Save and quit: Press ESC then :wq

3. Initialize Terraform:

terraform init

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ vim main.tf
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

4. Show contents of .terraform.lock.hcl:

cat .terraform.lock.hcl

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat .terraform.lock.hcl
# This file is maintained automatically by "terraform init".
# Manual edits may be lost in future updates.

provider "registry.terraform.io/hashicorp/aws" {
  version = "6.27.0"
  hashes = [
    "h1:bixp2PSsP5ZGBczGCxcSDn6lF5QFLUXlNroq9cdab4=",
    "zh:177a24b806c72e8484b5cabc93b2b38e3d770ae6f745a998b54d6619fd0e8129",
    "zh:4ac4a85c14fb868a3306b542e6a56c10bd6c6d5a67bc0c9b8f6a9060cf5f3be7",
    "zh:552652185bc85c8ba1da1d65dea47c454728a5c6839c458b6dcd3ce71c19ccfc",
    "zh:60284b8172d09aee91eae0856f09855eaf040ce3a58d6933602ae17c53f8ed04",
    "zh:6be38d156756ca61fb8e7c752cc5d769cd709686700ac4b230f40a6e95b5dbc9",
    "zh:7a409138fae4ef42e3a637e37cb9efedf96459e28a3c764fc4e855e8db9a7485",
    "zh:8070cf5224ed1ed3a3e9a59f7c30ff88b071c7567165275d477c1738a56c064",
    "zh:894439ef340a9a79f69cd759e27ad11c7826adeca27be1b1ca82b3c9702fa300",
    "zh:89d035eebf08a97c89374ff06040955ddc09f275ecca609d0c9d58d149bef5cf",
    "zh:985b1145d724fc1f38369099e4a5087141885740fd6c0b1dbc492171e73c2e49",
    "zh:9b12af85486a96aedd8d7984b0ff811a4b42e3d88dad1a3fb4c0b580d04fa425",
    "zh:a80b47ae8d1475201c86bd94a5dcb9dd4da5e8b73102a90820b68b66b76d50fd",
    "zh:d3395be1556210f82199b9166a6b2e677cee9c4b67e96e63f6c3a98325ad7ab0",
    "zh:db0b869d09657f6f1e4110b56093c5fcd9dbdd97c020db1e577b239c0adcbce",
    "zh:ffc72e680370ae7c21f9bd3082c6317730df805c6797427839a6b6b7e9a26a01",
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

5. Show contents of .terraform/ directory:

ls .terraform/

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ ls .terraform/
providers
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

Task 3 — VPC/Subnet Creation, Initialization, Verification

Goal: Use Terraform to create VPC & Subnet resources, verify via CLI.

1. Edit main.tf to add:

```
resource "aws_vpc" "development_vpc" {

  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {

  vpc_id   = aws_vpc.development_vpc.id

  cidr_block = "10.0.10.0/24"

  availability_zone = "me-central-1a"
}
```

```

provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]}
resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"

  tags = {
    Name = "development-vpc"
  }
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id      = aws_vpc.development_vpc.id
  cidr_block  = "10.0.10.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "dev-subnet-1"
  }
}
~
~

```

2. Use vim main.tf, edit and insert code.

Run:

terraform apply

```

@23-22411-087-sudo → /workspaces/exam_prep (main) $ terraform apply
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "dev-subnet-1"
  }
  + tags_all                           = {
    + "Name" = "dev-subnet-1"
  }
  + vpc_id                             = (known after apply)
}

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                         = "10.0.0.0/16"
  + default_network_acl_id             = (known after apply)
  + default_route_table_id            = (known after apply)
  + default_security_group_id         = (known after apply)
  + dhcp_options_id                   = (known after apply)
  + enable_dns_hostnames               = (known after apply)
  + enable_dns_support                 = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                 = (known after apply)
  + instance_tenancy                   = "default"
  + ipv6_association_id                = (known after apply)
  + ipv6_cidr_block                    = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id                = (known after apply)
  + owner_id                           = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "development-vpc"
  }
  + tags_all                           = {

```

```
+ resource "aws_vpc" "development_vpc" {
+   arn                                = (known after apply)
+   cidr_block                         = "10.0.0.0/16"
+   default_network_acl_id            = (known after apply)
+   default_route_table_id            = (known after apply)
+   default_security_group_id         = (known after apply)
+   dhcp_options_id                   = (known after apply)
+   enable_dns_hostnames               = (known after apply)
+   enable_dns_support                 = true
+   enable_network_address_usage_metrics = (known after apply)
+   id                                 = (known after apply)
+   instance_tenancy                   = "default"
+   ipv6_association_id                = (known after apply)
+   ipv6_cidr_block                    = (known after apply)
+   ipv6_cidr_block_network_border_group = (known after apply)
+   main_route_table_id                = (known after apply)
+   owner_id                           = (known after apply)
+   region                             = "me-central-1"
+   tags                               = {
+     "Name" = "development-vpc"
+   }
+   tags_all                           = {
+     "Name" = "development-vpc"
+   }
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.development_vpc: Creating...

aws_vpc.development_vpc: Creation complete after 2s [id=vpc-0323e67344d3db399]

aws_subnet.dev_subnet_1: Creating...

aws_subnet.dev_subnet_1: Creation complete after 1s [id=subnet-0188c3387205053ef]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

@23-22411-057-sudo → /workspaces/exam_prep (main) \$

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform show
# aws_subnet.dev_subnet_1:
resource "aws_subnet" "dev_subnet_1" {
  arn                                = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0188c3387205053ef"
  assign_ipv6_address_on_creation    = false
  availability_zone                  = "me-central-1a"
  availability_zone_id                = "mec1-az1"
  cidr_block                         = "10.0.10.0/24"
  customer_owned_ipv4_pool           = null
  enable_dns64                       = false
  enable_lni_at_device_index         = false
  enable_resource_name_dns_a_record_on_launch = false
  enable_resource_name_dns_aaaa_record_on_launch = false
  id                                 = "subnet-0188c3387205053ef"
  ipv6_cidr_block                    = null
  ipv6_cidr_block_association_id     = null
  ipv6_native                         = false
  map_customer_owned_ip_on_launch    = false
  map_public_ip_on_launch            = false
  outpost_arn                        = null
  owner_id                           = "359416636630"
  private_dns_hostname_type_on_launch = "ip-name"
  region                             = "me-central-1"
  tags                               = {
    "Name" = "dev-subnet-1"
  }
  tags_all                           = {
    "Name" = "dev-subnet-1"
  }
  vpc_id                             = "vpc-0323e67344d3db399"
}

# aws_vpc.development_vpc:
resource "aws_vpc" "development_vpc" {
  arn                                = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-0323e67344d3db399"
  assign_generated_ipv6_cidr_block   = false
  cidr_block                         = "10.0.0.0/16"
  default_network_acl_id             = "acl-0ea6ab17e937a5313"
  default_route_table_id             = "rtb-0ff685f2de83de46a"
  default_security_group_id          = "sg-0b2846e15fed998bf"
  dhcp_options_id                    = "dopt-07a746a94dc2af666"
  enable_dns_hostnames               = false
  enable_dns_support                  = true
  enable_network_address_usage_metrics = false
  id                                 = "vpc-0323e67344d3db399"
  instance_tenancy                   = "default"
  ipv6_association_id                = null
  ipv6_cidr_block                    = null
  ipv6_cidr_block_network_border_group = null
  ipv6_ipam_pool_id                  = null
  ipv6_netmask_length                 = 0
  main_route_table_id                = "rtb-0ff685f2de83de46a"
  owner_id                           = "359416636630"
  region                             = "me-central-1"
  tags                               = {
    "Name" = "development-vpc"
  }
}
```

Type yes when prompted.

successful apply and resources creation.

3. Verify resources using AWS CLI:

Run:

aws ec2 describe-subnets --filter "Name=subnet-id,Values=<subnet-id>"

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-vpcs \
--filters "Name=vpc-id,Values=vpc-0323e67344d3db399" \
--region me-central-1
{
  "Vpcs": [
    {
      "OwnerId": "359416636630",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0f461eed494ea42fc",
          "CidrBlock": "10.0.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": false,
      "Tags": [
        {
          "Key": "Name",
          "Value": "development-vpc"
        }
      ],
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-0323e67344d3db399",
      "State": "available",
      "CidrBlock": "10.0.0.0/16",
      "DhcpOptionsId": "dopt-07a746a94dc2af666"
    }
  ]
}
```

@23-22411-057-sudo → /workspaces/exam_prep (main) \$

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-subnets \
--filters "Name=subnet-id,Values=subnet-0188c3387205053ef" \
--region me-central-1
{
  "Subnets": [
    {
      "AvailabilityZoneId": "mec1-az1",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "359416636630",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "Tags": [
        {
          "Key": "Name",
          "Value": "dev-subnet-1"
        }
      ],
      "SubnetArn": "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0188c3387205053ef",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-0188c3387205053ef",
      "State": "available",
      "VpcId": "vpc-0323e67344d3db399",
      "CidrBlock": "10.0.10.0/24",
      "AvailableIpAddressCount": 251,
      "AvailabilityZone": "me-central-1a",
      "DefaultForAz": false,
      "MapPublicIpOnLaunch": false
    }
  ]
}
```

@23-22411-057-sudo → /workspaces/exam_prep (main) \$ |

Run:

aws ec2 describe-vpcs --filter "Name=vpc-id,Values=<vpc-id>"

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-vpcs \
--filters "Name=vpc-id,Values=vpc-0323e67344d3db399" \
--region me-central-1
{
  "Vpcs": [
    {
      "OwnerId": "359416636630",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0f461eed494ea42fc",
          "CidrBlock": "10.0.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": false,
      "Tags": [
        {
          "Key": "Name",
          "Value": "development-vpc"
        }
      ],
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-0323e67344d3db399",
      "State": "available",
      "CidrBlock": "10.0.0.0/16",
      "DhcpOptionsId": "dopt-07a746a94dc2af666"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |

```

Task 4 — Data Source, Targeted Destroy, Tags

A. Data Source & Resource Creation

1. Add to main.tf:

```

data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id    = data.aws_vpc.existing_vpc.id
  cidr_block = "172.31.48.0/24"
  availability_zone = "me-central-1a"
}

```

Use vim to edit and insert code.

```

provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"

  tags = {
    Name = "development-vpc"
  }
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "dev-subnet-1"
  }
}

data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id            = data.aws_vpc.existing_vpc.id
  cidr_block        = "172.31.48.0/24"
  availability_zone = "me-central-1a"
}

```

2. Apply configuration:

terraform apply

Confirm only the new subnet is created.

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0323e67344d3db399]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0188c3387205053ef]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags_all                           = (known after apply)
  + vpc_id                             = "vpc-05627a462c11486de"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-0faaa21c97a7f69a6]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |

```

Save screenshot as: task4_terraform_apply_datasource_resource.png — terraform output of resource creation.

B. Targeted Destroy & Refresh

1. Destroy only one resource:

terraform destroy -target=aws_subnet.dev_subnet_1_existing

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform destroy -target=aws_subnet.dev_subnet_1_existing
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0faaa21c97a7f69a6]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be destroyed
resource "aws_subnet" "dev_subnet_1_existing" {
  arn                                = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0faaa21c97a7f69a6" -> null
  assign_ipv6_address_on_creation   = false -> null
  availability_zone                 = "me-central-1a" -> null
  availability_zone_id              = "mecl-az1" -> null
  cidr_block                        = "172.31.48.0/24" -> null
  enable_dns64                      = false -> null
  enable_init_device_index          = 0 -> null
  enable_resource_name_dns_a_record_on_launch = false -> null
  enable_resource_name_dns_aaaa_record_on_launch = false -> null
  id                                = "subnet-0faaa21c97a7f69a6" -> null
  ipv6_native                       = false -> null
  map_customer_owned_ip_on_launch   = false -> null
  map_public_ip_on_launch           = false -> null
  owner_id                          = "359416636630" -> null
  private_dns_hostname_type_on_launch = "ip-name" -> null
  region                            = "me-central-1" -> null
  tags                              = {} -> null
  tags_all                          = {} -> null
  vpc_id                            = "vpc-05627a462c11486de" -> null
}
# (n unchanged attributes hidden)

Plan: 0 to add, 0 to change, 1 to destroy.

Warning: Resource targeting is in effect

You are creating a plan with the -target option, which means that the result of this plan may not represent all of the changes requested by the current configuration.

The -target option is not for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes
aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-0faaa21c97a7f69a6]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s

Warning: Applied changes may be incomplete

The plan was created with the -target option in effect, so some changes requested in the configuration may have been ignored and the output values may not be fully updated. Run the following command to verify that no other changes are pending:
  terraform plan

Note that the -target option is not suitable for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Destroy complete! Resources: 1 destroyed.

```

3. Refresh state:

terraform refresh

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform refresh
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0323e67344d3db399]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0188c3387205053ef]
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |

```

4. Re-apply configuration:

terraform apply

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0323e67344d3db399]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0188c3387205053ef]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation   = false
  + availability_zone                 = "me-central-1a"
  + availability_zone_id              = (known after apply)
  + cidr_block                        = "172.31.48.0/24"
  + enable_dns64                      = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                = (known after apply)
  + ipv6_cidr_block_association_id    = (known after apply)
  + ipv6_native                       = false
  + map_public_ip_on_launch           = false
  + owner_id                          = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                            = "me-central-1"
  + tags                              = (known after apply)
  + tags_all                          = (known after apply)
  + vpc_id                            = "vpc-05627a462c11486de"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes
aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-07bc569e14fee742c]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |

```

5. Destroy all resources:

terraform destroy


```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform destroy
No changes. No objects need to be destroyed.

Either you have not created any objects yet or the existing objects were already deleted outside of Terraform.

Destroy complete! Resources: 0 destroyed.
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

6. Plan configuration:

terraform plan

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform plan
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-05627a462c11486de]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                      = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                       = false
  + map_public_ip_on_launch           = false
  + owner_id                          = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                            = "me-central-1"
  + tags                              = {
    + "Name" = "dev-subnet-1"
  }
  + tags_all                          = {
    + "Name" = "dev-subnet-1"
  }
  + vpc_id                            = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                      = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                       = false
  + map_public_ip_on_launch           = false
  + owner_id                          = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                            = "me-central-1"
  + tags_all                          = (known after apply)
  + vpc_id                            = "vpc-05627a462c11486de"
}

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                         = "10.0.0.0/16"
  + default_network_acl_id            = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id         = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + dhcp_options_id                  = (known after apply)
```

```
# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                         = "10.0.0.0/16"
  + default_network_acl_id            = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id         = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support                = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                = (known after apply)
  + instance_tenancy                  = "default"
  + ipv6_association_id               = (known after apply)
  + ipv6_cidr_block                   = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id              = (known after apply)
  + owner_id                          = (known after apply)
  + region                            = "me-central-1"
  + tags                              = {
    + "Name" = "development-vpc"
  }
  + tags_all                          = {
    + "Name" = "development-vpc"
  }
}

Plan: 3 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

7. Apply again:

terraform apply

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-05627a462c11486de]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                        = "10.0.10.0/24"
  + enable_dns64                      = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch           = false
  + owner_id                          = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                            = "me-central-1"
  + tags                              = {
    + "Name" = "dev-subnet-1"
  }
  + tags_all                          = {
    + "Name" = "dev-subnet-1"
  }
  + vpc_id                            = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                        = "172.31.48.0/24"
  + enable_dns64                      = false

```

```

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id            = (known after apply)
  + default_route_table_id            = (known after apply)
  + default_security_group_id         = (known after apply)
  + dhcp_options_id                   = (known after apply)
  + enable_dns_hostnames               = (known after apply)
  + enable_dns_support                 = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                = (known after apply)
  + instance_tenancy                   = "default"
  + ipv6_association_id                = (known after apply)
  + ipv6_cidr_block                    = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id                = (known after apply)
  + owner_id                          = (known after apply)
  + region                            = "me-central-1"
  + tags                              = {
    + "Name" = "development-vpc"
  }
  + tags_all                          = {
    + "Name" = "development-vpc"
  }
}

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.development_vpc: Creating...
aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-0d27097e720754c89]
aws_vpc.development_vpc: Creation complete after 1s [id=vpc-0cbacbcd6d27b5927]
aws_subnet.dev_subnet_1: Creating...
aws_subnet.dev_subnet_1: Creation complete after 1s [id=subnet-0a39b9f5d13beac27]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```

C. Tagging Resources

1. Modify main.tf to add tags:

```

resource "aws_vpc" "development_vpc" {

  cidr_block = "10.0.0.0/16"

  tags = {

    Name = "development"

    vpc_env = "dev"

  }
}

```

```
}  
  
resource "aws_subnet" "dev_subnet_1" {  
  vpc_id    = aws_vpc.development_vpc.id  
  cidr_block = "10.0.10.0/24"  
  availability_zone = "me-central-1a"  
  tags = {  
    Name: "subnet-1-dev"  
  }  
}  
  
resource "aws_subnet" "dev_subnet_1_existing" {  
  vpc_id    = data.aws_vpc.existing_vpc.id  
  cidr_block = "172.31.48.0/24"  
  availability_zone = "me-central-1a"  
  tags = {  
    Name: "subnet-1-default"  
  }  
}
```

```

provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"

  tags = {
    Name     = "development"
    vpc_env = "dev"
  }
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-dev"
  }
}

data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id            = data.aws_vpc.existing_vpc.id
  cidr_block        = "172.31.48.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-default"
  }
}

```

2. Use vim to update tags.

Run:

terraform refresh

terraform apply -auto-approve

```

$23-22411-057~$ sudo -> /workspaces/exam_prep (main) $ terraform refresh
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0cbacbcd6d27b5927]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0d27097e720754c89]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0a39b9f5d13beac27]
$23-22411-057~$ sudo -> /workspaces/exam_prep (main) $ terraform apply -auto-approve
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0cbacbcd6d27b5927]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0d27097e720754c89]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0a39b9f5d13beac27]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be updated in-place
~ resource "aws_subnet" "dev_subnet_1" {
  id           = "subnet-0a39b9f5d13beac27"
  ~ tags       = {
    ~ "Name" = "dev-subnet-1" -> "subnet-1-dev"
  }
  ~ tags_all   = {
    ~ "Name" = "dev-subnet-1" -> "subnet-1-dev"
  }
  # (20 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be updated in-place
~ resource "aws_subnet" "dev_subnet_1_existing" {
  id           = "subnet-0d27097e720754c89"
  ~ tags       = {
    + "Name" = "subnet-1-default"
  }
  ~ tags_all   = {
    + "Name" = "subnet-1-default"
  }
  # (20 unchanged attributes hidden)
}

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id           = "vpc-0cbacbcd6d27b5927"

```

```

~ tags
~ "Name" = "dev-subnet-1" -> "subnet-1-dev" = {
}
~ tags_all
~ "Name" = "dev-subnet-1" -> "subnet-1-dev" = {
}
# (20 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be updated in-place
~ resource "aws_subnet" "dev_subnet_1_existing" {
  id = "subnet-0d27097e720754c89"
  ~ tags
  + "Name" = "subnet-1-default"
  ~ tags_all
  + "Name" = "subnet-1-default"
  # (20 unchanged attributes hidden)
}

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id = "vpc-0cbacbcd6d27b5927"
  ~ tags
  ~ "Name" = "development-vpc" -> "development"
  + "vpc_env" = "dev"
  ~ tags_all
  ~ "Name" = "development-vpc" -> "development"
  + "vpc_env" = "dev"
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 3 to change, 0 to destroy.
aws_subnet.dev_subnet_1_existing: Modifying... [id=subnet-0d27097e720754c89]
aws_vpc.development_vpc: Modifying... [id=vpc-0cbacbcd6d27b5927]
aws_subnet.dev_subnet_1_existing: Modifications complete after 1s [id=subnet-0d27097e720754c89]
aws_vpc.development_vpc: Modifications complete after 1s [id=vpc-0cbacbcd6d27b5927]
aws_subnet.dev_subnet_1: Modifying... [id=subnet-0a39b9f5d13beac27]
aws_subnet.dev_subnet_1: Modifications complete after 0s [id=subnet-0a39b9f5d13beac27]

Apply complete! Resources: 0 added, 3 changed, 0 destroyed.
023-22411-057-sudo → /workspaces/exam_prep (main) $

```

3. Remove vpc_env = "dev" tag from development_vpc resource, re-plan/apply:

```

provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"

  tags = {
    Name = "development"
  }
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-dev"
  }
}

data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id            = data.aws_vpc.existing_vpc.id
  cidr_block        = "172.31.48.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-default"
  }
}

```



```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform plan
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0cbacbcd6d27b5927]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0d27097e720754c89]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0a39b9f5d13beac27]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  ~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-0cbacbcd6d27b5927"
  ~ tags      = {
    ~ "Name" = "development"
    ~ "vpc_env" = "dev" -> null
  }
  ~ tags_all  = {
    ~ "vpc_env" = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

Task 5 — State File Inspection & Terraform State Commands

1. Destroy all resources:

terraform destroy

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform destroy
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0cbacbcd6d27b5927]
data.aws_vpc.existing_vpc: Read complete after 2s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0d27097e720754c89]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0a39b9f5d13beac27]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be destroyed
- resource "aws_subnet" "dev_subnet_1" {
  ~ arn              = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0a39b9f5d13beac27" -> null
  ~ assign_ipv6_address_on_creation = false -> null
  ~ availability_zone = "me-central-1a" -> null
  ~ availability_zone_id = "mec1-az1" -> null
  ~ cidr_block        = "10.0.10.0/24" -> null
  ~ enable_dns64       = false -> null
  ~ enable_lni_at_device_index = 0 -> null
  ~ enable_resource_name_dns_a_record_on_launch = false -> null
  ~ enable_resource_name_dns_aaaa_record_on_launch = false -> null
  ~ id                = "subnet-0a39b9f5d13beac27" -> null
  ~ ipv6_native        = false -> null
  ~ map_customer_owned_ip_on_launch = false -> null
  ~ map_public_ip_on_launch = false -> null
  ~ owner_id           = "359416636630" -> null
  ~ private_dns_hostname_type_on_launch = "ip-name" -> null
  ~ region             = "me-central-1" -> null
  ~ tags              = {
    ~ "Name" = "subnet-1-dev"
  } -> null
  ~ tags_all          = {
    ~ "Name" = "subnet-1-dev"
  } -> null
  ~ vpc_id            = "vpc-0cbacbcd6d27b5927" -> null
  # (4 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be destroyed
```

```
# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  ~ arn              = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0d27097e720754c89" -> null
  ~ assign_ipv6_address_on_creation = false -> null
  ~ availability_zone = "me-central-1a" -> null
  ~ availability_zone_id = "mec1-az1" -> null
  ~ cidr_block        = "172.31.48.0/24" -> null
  ~ enable_dns64       = false -> null
  ~ enable_lni_at_device_index = 0 -> null
  ~ enable_resource_name_dns_a_record_on_launch = false -> null
  ~ enable_resource_name_dns_aaaa_record_on_launch = false -> null
  ~ id                = "subnet-0d27097e720754c89" -> null
  ~ ipv6_native        = false -> null
  ~ map_customer_owned_ip_on_launch = false -> null
  ~ map_public_ip_on_launch = false -> null
  ~ owner_id           = "359416636630" -> null
  ~ private_dns_hostname_type_on_launch = "ip-name" -> null
  ~ region             = "me-central-1" -> null
  ~ tags              = {
    ~ "Name" = "subnet-1-default"
  } -> null
  ~ tags_all          = {
    ~ "Name" = "subnet-1-default"
  } -> null
  ~ vpc_id            = "vpc-05627a462c11486de" -> null
  # (4 unchanged attributes hidden)
}

# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  ~ arn              = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-0cbacbcd6d27b5927" -> null
  ~ assign_generated_ipv6_cidr_block = false -> null
  ~ cidr_block        = "10.0.0.0/16" -> null
  ~ default_network_acl_id = "acl-0b9fe5a45fd73c81c" -> null
  ~ default_route_table_id = "rtb-018ccf033b4a73a62" -> null
  ~ default_security_group_id = "sg-00bfd12db0057fb52" -> null
  ~ dhcp_options_id = "dopt-07a746a94dc2af666" -> null
  ~ enable_dns_hostnames = false -> null
  ~ enable_dns_support = true -> null
  ~ enable_network_address_usage_metrics = false -> null
  ~ id                = "vpc-0cbacbcd6d27b5927" -> null
```

```

- default_network_acl_id           = "acl-0b9fe5a45fd73c81c" -> null
- default_route_table_id          = "rtb-018ccf033b4a73a62" -> null
- default_security_group_id       = "sg-00bfd12db0057fb52" -> null
- dhcp_options_id                 = "dopt-07a746a94dc2af666" -> null
- enable_dns_hostnames            = false -> null
- enable_dns_support              = true -> null
- enable_network_address_usage_metrics = false -> null
- id                              = "vpc-0cbacbcd6d27b5927" -> null
- instance_tenancy                = "default" -> null
- ipv6_netmask_length             = 0 -> null
- main_route_table_id             = "rtb-018ccf033b4a73a62" -> null
- owner_id                        = "359416636630" -> null
- region                          = "me-central-1" -> null
- tags                            = {
  - "Name"      = "development"
  - "vpc_env"   = "dev"
} -> null
- tags_all                        = {
  - "Name"      = "development"
  - "vpc_env"   = "dev"
} -> null
# (4 unchanged attributes hidden)
}

```

Plan: 0 to add, 0 to change, 3 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```

aws_subnet.dev_subnet_1: Destroying... [id=subnet-0a39b9f5d13beac27]
aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-0d27097e720754c89]
aws_subnet.dev_subnet_1: Destruction complete after 1s
aws_vpc.development_vpc: Destroying... [id=vpc-0cbacbcd6d27b5927]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s
aws_vpc.development_vpc: Destruction complete after 1s

```

Destroy complete! Resources: 3 destroyed.

@23-22411-057-sudo → /workspaces/exam_prep (main) \$

2. Inspect state files after destroy:

Run:

cat terraform.tfstate

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 28,
  "lineage": "596b4b95-c7cb-3029-2221-188db0f3e360",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```

Run:

cat terraform.tfstate.backup


```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-05627a462c11486de]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "subnet-1-dev"
  }
  + tags_all                           = {
    + "Name" = "subnet-1-dev"
  }
  + vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "subnet-1-default"
  }
}

```

```

+ region                             = "me-central-1"
+ tags                               = {
  + "Name" = "subnet-1-default"
}
+ tags_all                           = {
  + "Name" = "subnet-1-default"
}
+ vpc_id                             = "vpc-05627a462c11486de"
}

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                         = "10.0.0.0/16"
  + default_network_acl_id             = (known after apply)
  + default_route_table_id            = (known after apply)
  + default_security_group_id         = (known after apply)
  + dhcp_options_id                   = (known after apply)
  + enable_dns_hostnames               = (known after apply)
  + enable_dns_support                 = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                 = (known after apply)
  + instance_tenancy                   = "default"
  + ipv6_association_id                = (known after apply)
  + ipv6_cidr_block                    = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id                = (known after apply)
  + owner_id                           = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "development"
  }
  + tags_all                           = {
    + "Name" = "development"
  }
}

```

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```

aws_vpc.development_vpc: Creating...
aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-0151fa6d628379d21]
aws_vpc.development_vpc: Creation complete after 2s [id=vpc-075d776e3de3d2b2a]
aws_subnet.dev_subnet_1: Creating...
aws_subnet.dev_subnet_1: Creation complete after 0s [id=subnet-09307b901117984d4]

```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

@23-22411-057-sudo → /workspaces/exam_prep (main) \$

4. View state files:

Run:

cat terraform.tfstate

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 32,
  "lineage": "596b4b95-c7cb-3029-2221-188db0f3e360",
  "outputs": {},
  "resources": [
    {
      "mode": "data",
      "type": "aws_vpc",
      "name": "existing_vpc",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 0,
          "attributes": {
            "arn": "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-05627a462c11486de",
            "cidr_block": "172.31.0.0/16",
            "cidr_block_associations": [
              {
                "association_id": "vpc-cidr-assoc-0185d7fd697e471",
                "cidr_block": "172.31.0.0/16",
                "state": "associated"
              }
            ],
            "default": true,
            "dhcp_options_id": "dopt-07a746a94dc2af666",
            "enable_dns_hostnames": true,
            "enable_dns_support": true,
            "enable_network_address_usage_metrics": false,
            "filter": null,
            "id": "vpc-05627a462c11486de",
            "instance_tenancy": "default",
            "ipv6_association_id": "",
            "ipv6_cidr_block": "",
            "main_route_table_id": "rtb-04868885e737881c0",
            "owner_id": "359416636630",
            "region": "me-central-1",
            "state": null,
            "tags": {},
            "timeouts": null
          },
          "sensitive_attributes": [],
          "identity_schema_version": 0
        }
      ]
    },
    {
      "mode": "managed",
      "type": "aws_subnet",
      "name": "dev_subnet_1",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 1,
```

Run:

cat terraform.tfstate.backup

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat terraform.tfstate.backup
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 28,
  "lineage": "596b4b95-c7cb-3029-2221-188db0f3e360",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

5. List resources:

terraform state list

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform state list
data.aws_vpc.existing_vpc
aws_subnet.dev_subnet_1
aws_subnet.dev_subnet_1_existing
aws_vpc.development_vpc
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```


6. Show full attributes:

terraform state show <resource-name>

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform state show aws_vpc.development_vpc
# aws_vpc.development_vpc:
resource "aws_vpc" "development_vpc" {
  arn                                = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-075d776e3de3d2b2a"
  assign_generated_ipv6_cidr_block   = false
  cidr_block                         = "10.0.0.0/16"
  default_network_acl_id            = "acl-0426b100e2e6ec0c6"
  default_route_table_id            = "rtb-0a8ad2ea0665b6e96"
  default_security_group_id         = "sg-0917ff8a79b318c5d"
  dhcp_options_id                   = "dopt-07a746a94dc2af666"
  enable_dns_hostnames               = false
  enable_dns_support                 = true
  enable_network_address_usage_metrics = false
  id                                = "vpc-075d776e3de3d2b2a"
  instance_tenancy                   = "default"
  ipv6_association_id               = null
  ipv6_cidr_block                   = null
  ipv6_cidr_block_network_border_group = null
  ipv6_ipam_pool_id                 = null
  ipv6_netmask_length               = 0
  main_route_table_id               = "rtb-0a8ad2ea0665b6e96"
  owner_id                          = "359416636630"
  region                            = "me-central-1"
  tags                               = {
    "Name" = "development"
  }
  tags_all                           = {
    "Name" = "development"
  }
}
```

Task 6 — Terraform Outputs & Attributes Reporting

1. Add outputs in main.tf:

```
output "dev-vpc-id" {
  value = aws_vpc.development_vpc.id
}
output "dev-subnet-id" {
  value = aws_subnet.dev_subnet_1.id
}
output "dev-vpc-arn" {
  value = aws_vpc.development_vpc.arn
}
output "dev-subnet-arn" {
  value = aws_subnet.dev_subnet_1.arn
}
```

```

resource "aws_subnet" "dev_subnet_1" {
  vpc_id          = aws_vpc.development_vpc.id
  cidr_block      = "10.0.10.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-dev"
  }
}

data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id          = data.aws_vpc.existing_vpc.id
  cidr_block      = "172.31.48.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-default"
  }
}

output "dev-vpc-id" {
  value = aws_vpc.development_vpc.id
}
output "dev-subnet-id" {
  value = aws_subnet.dev_subnet_1.id
}
output "dev-vpc-arn" {
  value = aws_vpc.development_vpc.arn
}
output "dev-subnet-arn" {
  value = aws_subnet.dev_subnet_1.arn
}

```

"main.tf" 49L, 977B

2. Return ALL of the following output attributes from your VPC and Subnet resources:

a) **cidr_block**

b) **region**

c) **tags.Name**

d) **tags_all**

Expand output section in main.tf accordingly:

```

output "dev-vpc-cidr_block" {
  value = aws_vpc.development_vpc.cidr_block
}
output "dev-vpc-region" {
  value = aws_vpc.development_vpc.region
}
output "dev-vpc-tags_name" {
  value = aws_vpc.development_vpc.tags["Name"]
}
output "dev-vpc-tags_all" {
  value = aws_vpc.development_vpc.tags_all
}
output "dev-subnet-cidr_block" {
  value = aws_subnet.dev_subnet_1.cidr_block
}

```

```

}output "dev-subnet-region" {

  value = aws_subnet.dev_subnet_1.availability_zone
}output "dev-subnet-tags_name" {

  value = aws_subnet.dev_subnet_1.tags["Name"]
}output "dev-subnet-tags_all" {

  value = aws_subnet.dev_subnet_1.tags_all
}

```

Run terraform apply and record outputs.

```

tags = {
  Name = "subnet-1-default"
}
}
output "dev-vpc-id" {
  value = aws_vpc.development_vpc.id
}
output "dev-subnet-id" {
  value = aws_subnet.dev_subnet_1.id
}
output "dev-vpc-arn" {
  value = aws_vpc.development_vpc.arn
}
output "dev-subnet-arn" {
  value = aws_subnet.dev_subnet_1.arn
}
output "dev-vpc-cidr_block" {
  value = aws_vpc.development_vpc.cidr_block
}
output "dev-vpc-region" {
  value = aws_vpc.development_vpc.region
}
output "dev-vpc-tags_name" {
  value = aws_vpc.development_vpc.tags["Name"]
}
output "dev-vpc-tags_all" {
  value = aws_vpc.development_vpc.tags_all
}
output "dev-subnet-cidr_block" {
  value = aws_subnet.dev_subnet_1.cidr_block
}
output "dev-subnet-region" {
  value = aws_subnet.dev_subnet_1.availability_zone
}
output "dev-subnet-tags_name" {
  value = aws_subnet.dev_subnet_1.tags["Name"]
}
output "dev-subnet-tags_all" {
  value = aws_subnet.dev_subnet_1.tags_all
}
-- INSERT --

```

```

823-22411-057-sudo → /workspaces/exam_prep (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
823-22411-057-sudo → /workspaces/exam_prep (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-075d776e3de3d2b2a]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0151fa6d628379d21]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-09307b901117984d4]

Changes to Outputs:
+ dev-subnet-arn      = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-09307b901117984d4"
+ dev-subnet-cidr_block = "10.0.10.0/24"
+ dev-subnet-id       = "subnet-09307b901117984d4"
+ dev-subnet-region   = "me-central-1a"
+ dev-subnet-tags_all = {
+   Name = "subnet-1-dev"
}
+ dev-subnet-tags_name = "subnet-1-dev"
+ dev-vpc-arn          = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-075d776e3de3d2b2a"
+ dev-vpc-cidr_block   = "10.0.0.0/16"
+ dev-vpc-id           = "vpc-075d776e3de3d2b2a"
+ dev-vpc-region       = "me-central-1"
+ dev-vpc-tags_all     = {
+   Name = "development"
}
+ dev-vpc-tags_name    = "development"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

```

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

dev-subnet-arn = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-09307b901117984d4"
dev-subnet-cidr_block = "10.0.10.0/24"
dev-subnet-id = "subnet-09307b901117984d4"
dev-subnet-region = "me-central-1a"
dev-subnet-tags_all = tomap({
  "Name" = "subnet-1-dev"
})
dev-subnet-tags_name = "subnet-1-dev"
dev-vpc-arn = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-075d776e3de3d2b2a"
dev-vpc-cidr_block = "10.0.0.0/16"
dev-vpc-id = "vpc-075d776e3de3d2b2a"
dev-vpc-region = "me-central-1"
dev-vpc-tags_all = tomap({
  "Name" = "development"
})
dev-vpc-tags_name = "development"
823-22411-057-sudo → /workspaces/exam_prep (main) $ |

```

Cleanup — Delete Resources & State Verification

1. Destroy all resources:

terraform destroy

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform destroy
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-075d776e3de3d2b2a]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-05627a462c11486de]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0151fa6d628379d21]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-09307b901117984d4]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be destroyed
- resource "aws_subnet" "dev_subnet_1" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-09307b901117984d4" -> null
  - assign_ipv6_address_on_creation = false -> null
  - availability_zone = "me-central-1a" -> null
  - availability_zone_id = "mecl-az1" -> null
  - cidr_block = "10.0.10.0/24" -> null
  - enable_dns64 = false -> null
  - enable_lni_at_device_index = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id = "subnet-09307b901117984d4" -> null
  - ipv6_native = false -> null
  - map_customer_owned_ip_on_launch = false -> null
  - map_public_ip_on_launch = false -> null
  - owner_id = "359416636630" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region = "me-central-1" -> null
  - tags = {
    - "Name" = "subnet-1-dev"
  } -> null
  - tags_all = {
    - "Name" = "subnet-1-dev"
  } -> null
  - vpc_id = "vpc-075d776e3de3d2b2a" -> null
  # (4 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be destroyed

```

```

# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0151fa6d628379d21" -> null
  - assign_ipv6_address_on_creation = false -> null
  - availability_zone = "me-central-1a" -> null
  - availability_zone_id = "mecl-az1" -> null
  - cidr_block = "172.31.48.0/24" -> null
  - enable_dns64 = false -> null
  - enable_lni_at_device_index = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id = "subnet-0151fa6d628379d21" -> null
  - ipv6_native = false -> null
  - map_customer_owned_ip_on_launch = false -> null
  - map_public_ip_on_launch = false -> null
  - owner_id = "359416636630" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region = "me-central-1" -> null
  - tags = {
    - "Name" = "subnet-1-default"
  } -> null
  - tags_all = {
    - "Name" = "subnet-1-default"
  } -> null
  - vpc_id = "vpc-05627a462c11486de" -> null
  # (4 unchanged attributes hidden)
}

# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-075d776e3de3d2b2a" -> null
  - assign_generated_ipv6_cidr_block = false -> null
  - cidr_block = "10.0.0.0/16" -> null
  - default_network_acl_id = "acl-08d26b108e2e6ec0c6" -> null
  - default_route_table_id = "rtb-0a8ad2ea0665b6e96" -> null
  - default_security_group_id = "sg-0917ff8a79b318c5d" -> null
  - dhcp_options_id = "dopt-07a746a94dc2af666" -> null
  - enable_dns_hostnames = false -> null
  - enable_dns_support = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id = "vpc-075d776e3de3d2b2a" -> null
}

```

Plan: 0 to add, 0 to change, 3 to destroy.

```

Changes to Outputs:
- dev-subnet-arn = "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-09307b901117984d4" -> null
- dev-subnet-cidr_block = "10.0.10.0/24" -> null
- dev-subnet-id = "subnet-09307b901117984d4" -> null
- dev-subnet-region = "me-central-1a" -> null
- dev-subnet-tags_all = {
  - Name = "subnet-1-dev"
} -> null
- dev-subnet-tags_name = "subnet-1-dev" -> null
- dev-vpc-arn = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-075d776e3de3d2b2a" -> null
- dev-vpc-cidr_block = "10.0.0.0/16" -> null
- dev-vpc-id = "vpc-075d776e3de3d2b2a" -> null
- dev-vpc-region = "me-central-1" -> null
- dev-vpc-tags_all = {
  - Name = "development"
} -> null
- dev-vpc-tags_name = "development" -> null

```

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```

aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-0151fa6d628379d21]
aws_subnet.dev_subnet_1: Destroying... [id=subnet-09307b901117984d4]
aws_subnet.dev_subnet_1: Destruction complete after 0s
aws_vpc.development_vpc: Destroying... [id=vpc-075d776e3de3d2b2a]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s
aws_vpc.development_vpc: Destruction complete after 1s

```

```

Destroy complete! Resources: 3 destroyed.
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```

2. Inspect state files:

cat terraform.tfstate

cat terraform.tfstate.backup

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 38,
  "lineage": "596b4b95-c7cb-3029-2221-188db0f3e360",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat terraform.tfstate.backup
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 33,
  "lineage": "596b4b95-c7cb-3029-2221-188db0f3e360",
  "outputs": {
    "dev-subnet-arn": {
      "value": "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-09307b901117984d4",
      "type": "string"
    },
    "dev-subnet-cidr_block": {
      "value": "10.0.10.0/24",
      "type": "string"
    },
    "dev-subnet-id": {
      "value": "subnet-09307b901117984d4",
      "type": "string"
    },
    "dev-subnet-region": {
      "value": "me-central-1a",
      "type": "string"
    },
    "dev-subnet-tags_all": {
      "value": {
        "Name": "subnet-1-dev"
      },
      "type": [
        "map",
        "string"
      ]
    }
  }
}
```

```
"resources": [
  {
    "mode": "data",
    "type": "aws_vpc",
    "name": "existing_vpc",
    "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
    "instances": [
      {
        "schema_version": 0,
        "attributes": {
          "arn": "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-85627a462c11486de",
          "cidr_block": "172.31.0.0/16",
          "cidr_block_associations": [
            {
              "association_id": "vpc-cidr-assoc-0185d7fdfa697e471",
              "cidr_block": "172.31.0.0/16",
              "state": "associated"
            }
          ],
          "default": true,
          "dhcp_options_id": "dopt-07a746a94dc2af666",
          "enable_dns_hostnames": true,
          "enable_dns_support": true,
          "enable_network_address_usage_metrics": false,
          "filter": null,
          "id": "vpc-85627a462c11486de",
          "instance_tenancy": "default",
          "ipv6_association_id": "",
          "ipv6_cidr_block": "",
          "main_route_table_id": "rtb-04868885e737881c8",
          "owner_id": "359416636630",
          "region": "me-central-1",
          "state": null,
          "tags": {},
          "timeouts": null
        },
        "sensitive_attributes": [],
        "identity_schema_version": 0
      }
    ]
  },
  {
    "mode": "managed",
    "type": "aws_subnet",
    "name": "dev_subnet_1",
    "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
    "instances": [
      {
        "schema_version": 1,
        "attributes": {
          "arn": "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-09307b901117984d4",
          "assign_ipv6_address_on_creation": false,
          "availability_zone": "me-central-1a",
          "availability_zone_id": "mecl-az1",
          "cidr_block": "10.0.10.0/24",
          "customer_owned_ipv4_pool": "",

```

```
{
  "mode": "managed",
  "type": "aws_subnet",
  "name": "dev_subnet_1_existing",
  "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
  "instances": [
    {
      "schema_version": 1,
      "attributes": {
        "arn": "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0151fa6d628379d21",
        "assign_ipv6_address_on_creation": false,
        "availability_zone": "me-central-1a",
        "availability_zone_id": "mecl-a21",
        "cidr_block": "172.31.48.0/24",
        "customer_owned_ipv4_pool": "",
        "enable_dns_sd": false,
        "enable_lni_at_device_index": 0,
        "enable_resource_name_dns_a_record_on_launch": false,
        "enable_resource_name_dns_aaaa_record_on_launch": false,
        "id": "subnet-0151fa6d628379d21",
        "ipv6_cidr_block": "",
        "ipv6_cidr_block_association_id": "",
        "ipv6_native": false,
        "map_customer_owned_ip_on_launch": false,
        "map_public_ip_on_launch": false,
        "outpost_arn": "",
        "owner_id": "359416636630",
        "private_dns_host_name_type_on_launch": "ip-name",
        "region": "me-central-1",
        "tags": {
          "Name": "subnet-1-default"
        }
      },
      "tags_all": {
        "Name": "subnet-1-default"
      },
      "timeouts": null,
      "vpc_id": "vpc-086274a462c11486de"
    },
    {
      "sensitive_attributes": [],
      "identity_schema_version": 0,
      "identity": {
        "account_id": "359416636630",
        "id": "subnet-0151fa6d628379d21",
        "region": "me-central-1"
      },
      "private": "eyJ3MmJmYjczMCllV2FhLTExZTYtOGVhOGEzNDMxN2QzMW8yZiAiOnsiaXNlYXRlIjo2MDAwMDAwMDAwInRlbGVhbnQ1IGRTIiwudDAwMDAwMDAwMH0iaWlnaGVlYVY9ZDZkZWduIjoISj9",
      "dependencies": [
        "data.aws_vpc.existing_vpc"
      ]
    }
  ]
},
{
  "mode": "managed",
  "type": "aws_vpc",
  "name": "development_vpc",
```

```

"mode": "managed",
"type": "aws_vpc",
"name": "development_vpc",
"provider": ["registry.terraform.io/hashicorp/aws"],
"instances": [
{
"schema_version": 1,
"attributes": {
"arn": "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-075d776e3de3d2b2a",
"assign_generated_ipv6_cidr_block": false,
"cidr_block": "10.0.0.0/16",
"default_network_acl_id": "acl-0426b108e2e6ec0c6",
"default_route_table_id": "rtb-0a8ad2ea0665b6e96",
"default_security_group_id": "sg-0917ff8a79b318c5d",
"dhcp_options_id": "dopt-07a746a94dc2af666",
"enable_dns_hostnames": false,
"enable_dns_support": true,
"enable_network_address_usage_metrics": false,
"id": "vpc-075d776e3de3d2b2a",
"instance_tenancy": "default",
"ipv4_ipam_pool_id": null,
"ipv4_netmask_length": null,
"ipv6_association_id": "",
"ipv6_cidr_block": "",
"ipv6_cidr_block_network_border_group": "",
"ipv6_ipam_pool_id": "",
"ipv6_netmask_length": 0,
"main_route_table_id": "rtb-0a8ad2ea0665b6e96",
"owner_id": "359416636630",
"region": "me-central-1",
"tags": {
"Name": "development"
},
"tags_all": {
"Name": "development"
}
},
"sensitive_attributes": [],
"identity_schema_version": 0,
"identity": {
"account_id": "359416636630",
"id": "vpc-075d776e3de3d2b2a",
"region": "me-central-1"
},
"private": "eyJzY2h1bWFnZmVyc2lvbWl6IjEifQ=="
}
]
},
"check_results": null
}

```

3. Reapply then compare state and backup files to see differences.

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ terraform apply -auto-approve
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-05627a462c11486de]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "subnet-1-dev"
  }
  + tags_all                           = {
    + "Name" = "subnet-1-dev"
  }
  + vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "subnet-1-default"
  }
  + tags_all                           = {

```

Compare state and Backup Files:

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 42,
  "lineage": "596b4b95-c7cb-3029-2221-188db0f3e360",
  "outputs": {
    "dev-subnet-arn": {
      "value": "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-074fbd6f85279d82f",
      "type": "string"
    },
    "dev-subnet-cidr_block": {
      "value": "10.0.10.0/24",
      "type": "string"
    },
    "dev-subnet-id": {
      "value": "subnet-074fbd6f85279d82f",
      "type": "string"
    },
    "dev-subnet-region": {
      "value": "me-central-1a",
      "type": "string"
    },
    "dev-subnet-tags_all": {
      "value": {
        "Name": "subnet-1-dev"
      },
      "type": [
        "map",
        "string"
      ]
    },
    "dev-subnet-tags_name": {
      "value": "subnet-1-dev",
      "type": "string"
    },
    "dev-vpc-arn": {
      "value": "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-0a08a092b519085e9",
      "type": "string"
    },
    "dev-vpc-cidr_block": {
      "value": "10.0.0.0/16",
      "type": "string"
    },
    "dev-vpc-id": {

```

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat terraform.tfstate.backup
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 38,
  "lineage": "596b4b95-c7cb-3029-2221-188db0f3e360",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```